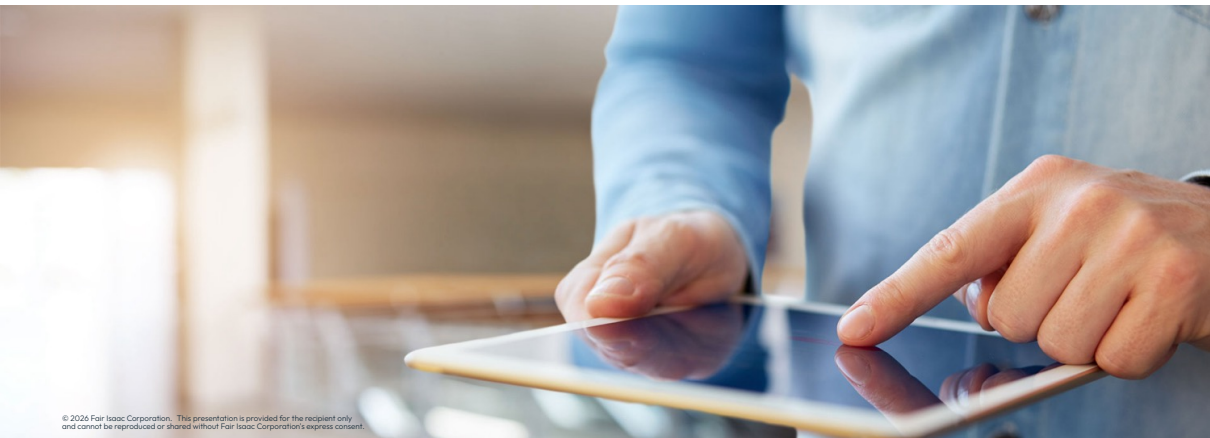


PWL functions



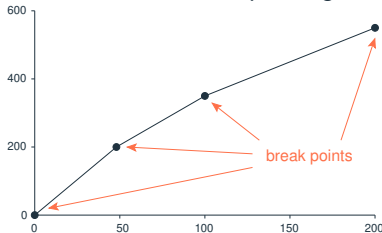


Piecewise linear (PWL) functions

Piecewise linear (PWL) functions

- Piecewise linear constraints define a variable as a **piecewise linear function of another variable**:

- Also used to **model stepwise functions** or to **approximate nonlinear functions**
- Example for discounts on unit costs depending on the quantity of items bought:



- First 50 items: $COST_1 = \$4$ each
 - Next 50 items: $COST_2 = \$3$ each
 - Then, up to 200: $COST_3 = \$2$ each
- Quantity break points x_i : 0, 50, 100, 200
 - Cost break points y_i (= total cost of buying quantity x_i): 0, 200, 350, 550
 $y_i = COST_i \cdot (x_i - x_{i-1}) + y_{i-1}$ for $i = 1, 2, 3$

Piecewise linear (PWL) functions

- Piecewise linear functions can be intuitively added to a problem by using the `xp.pwl(dict)` method in constraints or objectives:
 - Receives a **dictionary** as argument that associates intervals with linear functions:
 - Dictionary has **tuples** of two elements as **keys** and linear expressions (or constants) as **values**
 - **Tuples specify the range** of the input variable for which the **expression is used as the function value**

Piecewise linear (PWL) functions

- Piecewise linear functions can be intuitively added to a problem by using the `xp.pwl(dict)` method in constraints or objectives:
 - Receives a **dictionary** as argument that associates intervals with linear functions:
 - Dictionary has **tuples** of two elements as **keys** and linear expressions (or constants) as **values**
 - **Tuples specify the range** of the input variable for which the **expression is used as the function value**
 - Modeling the previous example where y is a piecewise linear function of x :

```
x = p.addVariable(vartype=xp.integer, ub=200)
y = p.addVariable()
p.addConstraint(xp.pwl({(0, 50): 4*x,
                       (50, 100): 3*(x-50) + 200,
                       (100, 200): 2*(x-100) + 350})) == y)
```



Note: *The piecewise linear function is always univariate, i.e. there must always be only one input variable*

Piecewise linear (PWL) functions

- Piecewise linear functions can also be used as **components of expressions** in an optimization problem:

```
cons1 = y + 3*z**2 <= 3*xp.pwl(({(0, 1): x + 4, (1, 3): 1}))  
p.addConstraint(cons1)
```

Piecewise linear (PWL) functions

- Piecewise linear functions can also be used as **components of expressions** in an optimization problem:

```
cons1 = y + 3*z**2 <= 3*xp.pwl(({(0, 1): x + 4, (1, 3): 1}))  
p.addConstraint(cons1)
```

- Step functions** need a further specification if a variable does not appear in the values; in this case we must specify an additional key-value pair as `None: x` for that variable:

```
p.setObjective(xp.pwl(({(0, 1): 4, (1, 2): 1, (2, 3): 3, None: x}))
```

Piecewise linear (PWL) functions

- Piecewise linear functions can also be used as **components of expressions** in an optimization problem:

```
cons1 = y + 3*z**2 <= 3*xp.pwl(({(0, 1): x + 4, (1, 3): 1}))  
p.addConstraint(cons1)
```

- Step functions** need a further specification if a variable does not appear in the values; in this case we must specify an additional key-value pair as `None: x` for that variable:

```
p.setObjective(xp.pwl(({(0, 1): 4, (1, 2): 1, (2, 3): 3, None: x}))
```

- Discontinuities** in the function are allowed, for example:

```
xp.pwl(({(1, 2): 2*x + 4, (2, 3): x - 1}))
```

- Which is discontinuous at 2, the function value for $x=2$ will be either 8 or 1



Note: Check the *FICO® Xpress Optimizer reference manual* for more information on how to deal with discontinuous functions



Thank You

www.fico.com/optimization